Quadratics & Nonlinear Equations

ANSWER KEY

Solving quadratics by factoring

1 Solve $x^2 - 5x + 6 = 0$ by factoring.

Factor: $(x-2)(x-3) = 0$, so $x = 2$ or $x = 3$.

2 Solve $2x^2 + 7x - 4 = 0$ by factoring.

Factor: $(2x-1)(x+4) = 0$, so $x = \frac{1}{2}$ or $x = -4$.

Solving quadratics with the quadratic formula

3 Solve $x^2 + 4x - 1 = 0$ using the quadratic formula, giving exact answers.

$x = \frac{-4 \pm \sqrt{16+4}}{2} = \frac{-4 \pm \sqrt{20}}{2} = \frac{-4 \pm 2\sqrt{5}}{2} = -2 \pm \sqrt{5}$.

4 Use the discriminant to determine the number of real solutions of $3x^2 - 2x + 5 = 0$, without solving the equation.

Discriminant $= b^2 - 4ac = (-2)^2 - 4(3)(5) = 4 - 60 = -56$. Since the discriminant is negative, the equation has no real solutions (two complex solutions).

Completing the square and vertex form

5 Write $f(x) = x^2 - 6x + 11$ in vertex form by completing the square, and state the vertex.

$x^2 - 6x + 11 = (x^2 - 6x + 9) + 2 = (x-3)^2 + 2$. Vertex form: $f(x) = (x-3)^2 + 2$. Vertex: $(3, 2)$.

6 A parabola has vertex $(2, -5)$ and passes through $(4, -1)$. Find its equation in vertex form.

Vertex form: $y = a(x-2)^2 - 5$. Substitute $(4, -1)$: $-1 = a(4-2)^2 - 5$, so $-1 = 4a - 5$, giving $4a = 4$, $a = 1$. Equation: $y = (x-2)^2 - 5$.

Finding zeros, maximum, and minimum values

7 A ball's height is modeled by $h(t) = -16t^2 + 64t + 5$, where t is time in seconds. Find the maximum height and the time at which it occurs.

The maximum occurs at $t = -\frac{b}{2a} = -\frac{64}{2(-16)} = 2$ seconds. Maximum height: $h(2) = -16(4) + 64(2) + 5 = -64 + 128 + 5 = 69$ feet.

8 Find the zeros of $f(x) = x^2 - 9$, and interpret them as x-intercepts on a graph.

$x^2 - 9 = 0$, so $x^2 = 9$, giving $x = 3$ or $x = -3$. These are the two points where the parabola crosses the x-axis: $(3, 0)$ and $(-3, 0)$.

Systems with one linear and one quadratic equation

9 Solve the system: $y = x^2 - 2x - 3$ and $y = x - 1$.

Set equal: $x^2 - 2x - 3 = x - 1$. Subtract $x - 1$ from both sides: $x^2 - 3x - 2 = 0$. Using the quadratic formula: $x = \frac{3 \pm \sqrt{9 + 8}}{2} = \frac{3 \pm \sqrt{17}}{2}$. The corresponding y-values come from $y = x - 1$ for each x-value.

Exponential growth and decay

10 A population of bacteria triples every hour, starting with 200 bacteria. Write an exponential model and find the population after 4 hours.

Model: $P(t) = 200(3)^t$. After 4 hours: $P(4) = 200(3^4) = 200(81) = 16200$ bacteria.

11 A car worth 24000 dollars depreciates by 15 percent each year. Write an exponential model and find its value after 3 years, to the nearest dollar.

Model: $V(t) = 24000(0.85)^t$. After 3 years: $V(3) = 24000(0.85)^3 = 24000(0.614125) \approx 14739$ dollars.

Radical and rational equations

12 Solve $\sqrt{2x + 3} = 5$, and check your solution.

Square both sides: $2x + 3 = 25$, so $2x = 22$, giving $x = 11$. Check: $\sqrt{2(11) + 3} = \sqrt{25} = 5$ ✓.

13 Solve $\frac{3}{x - 2} = \frac{1}{x + 1}$, and state any restrictions on x.

Cross-multiply: $3(x + 1) = 1(x - 2)$, so $3x + 3 = x - 2$, giving $2x = -5$, $x = -2.5$. Restrictions: $x \neq 2$ and $x \neq -1$ (values that would make a denominator zero); since $x = -2.5$ avoids both, it is valid.

Nonlinear relationships in context

14 The area of a square garden is modeled by $A(s) = s^2$, where s is the side length in meters. If the area must be at most 144 square meters, find the maximum possible side length.

$s^2 \leq 144$, so $s \leq 12$ (taking the positive root since side length must be positive). The maximum side length is 12 meters.

Factoring special products

15 Factor $9x^2 - 25$ completely.

This is a difference of squares: $9x^2 - 25 = (3x - 5)(3x + 5)$.

16 Factor $x^2 + 10x + 25$ completely, and identify the special pattern used.

This is a perfect square trinomial: $x^2 + 10x + 25 = (x + 5)^2$.

Converting between standard, factored, and vertex forms

17 A parabola is given in factored form as $y = 2(x - 1)(x + 3)$. Convert this to standard form.

Expand: $2(x - 1)(x + 3) = 2(x^2 + 3x - x - 3) = 2(x^2 + 2x - 3) = 2x^2 + 4x - 6$.

18 A parabola has zeros at $x = 2$ and $x = 6$, and passes through $(4, -8)$. Find its equation in factored form.

Factored form: $y = a(x - 2)(x - 6)$. Substitute $(4, -8)$: $-8 = a(4 - 2)(4 - 6) = a(2)(-2) = -4a$, so $a = 2$. Equation: $y = 2(x - 2)(x - 6)$.

Quadratic word problems: projectile motion

19 A rocket is launched with height modeled by $h(t) = -16t^2 + 96t$, where t is time in seconds. Find when the rocket hits the ground, and find its maximum height.

The rocket hits the ground when $h(t) = 0$: $-16t^2 + 96t = 0$, so $-16t(t - 6) = 0$, giving $t = 0$ or $t = 6$ seconds (the rocket lands at $t = 6$). Maximum height occurs at $t = -\frac{96}{2(-16)} = 3$ seconds: $h(3) = -16(9) + 96(3) = -144 + 288 = 144$ feet.

Quadratic optimization in business contexts

20 A company's revenue is modeled by $R(x) = -5x^2 + 300x$, where x is the price per item in dollars. Find the price that maximizes revenue, and find the maximum revenue.

Maximum occurs at $x = -\frac{300}{2(-5)} = 30$ dollars. Maximum revenue: $R(30) = -5(900) + 300(30) = -4500 + 9000 = 4500$ dollars.

Determining the number of solutions graphically

21 A parabola $y = x^2 - 4x + 7$ and a line $y = 2x - 5$ are graphed together. Use the discriminant of the resulting equation to determine how many times the graphs intersect.

Set equal: $x^2 - 4x + 7 = 2x - 5$, so $x^2 - 6x + 12 = 0$. Discriminant: $(-6)^2 - 4(1)(12) = 36 - 48 = -12$. Since the discriminant is negative, there are no real solutions, meaning the parabola and line never intersect.